

This manual is to guide the users through how to use the `internsg.com_search.ipynb` to generate an Internsg job list.

1. Set up all essential tools

Before running the code, please follow the demo to complete the steps below to set up all necessary tools.

Visual Studio Code (VS Code)

This is a platform that we will use to run the code.

- Download VS Code (<https://code.visualstudio.com/download>) and open it.
- Inside the software, download Jupyter and Python in the Extensions view.
- Open the InternSG Job List folder and rename it to InternSG_Job_List.
- Enter the following command into the terminal to create a virtual environment:

```
python -m venv ./venv  
.\venv\Scripts\activate
```
- Click “Select Kernel” and “Python Environments” to choose the kernel called “venv”.
- Add a code cell. Type `print("Hello World!")` and run the code.
- You will see a pop up message. Click “Install” to install the ipykernel package for running the Jupyter notebook in VS Code.
- Type `pip install selenium` in the terminal to install Selenium.

Chrome Driver

Selenium and Chrome Driver are useful for scraping information from the Internet.

- Make sure your Chrome is up to date.
- Download Chrome Driver
(<https://googlechromelabs.github.io/chrome-for-testing/#stable>).
- If you are using Windows, copy the URL for the chromedriver win32 or win64.
- If you are using a Mac, copy the URL for the chromedriver mac-x64.
- Paste the URL in your browser, and a ZIP file will be downloaded automatically.
- Unzip the file and place the entire folder into your C drive.

2. Gain a basic understanding of the code

There are seven functions inside the first code cell. The introduction to each function is as follows:

- `create_chrome_driver(headless=True)`

This function is used to set up Chrome options. It will return a Chrome Driver in headless mode. This mode enables the fetching process to be faster.

- extract_contact_info(detail_soup)

Application	Email: careers@plexxie.com
Instructions	Interested candidates should submit their CV with headshot. Only shortlisted candidates will be notified. Thank you for applying!

This function is used to extract the contact information under the Application Instructions section of the job description page. If an email address is found in this section, the code will store the email (careers@plexxie.com). Additionally, if there are any links provided in this section, they will be stored for later use.

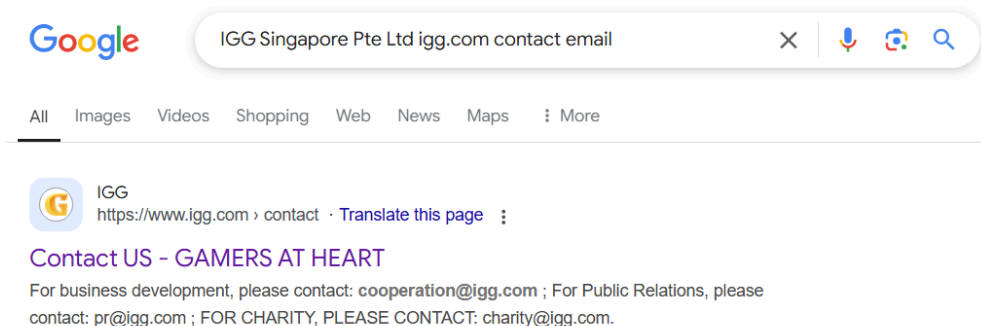
- extract_company_name(detail_soup)

The Plexxie Global Company Pte Ltd – Content Creator Intern

Company	The Plexxie Global Company Pte Ltd plexxie.com
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This function is used to extract the company name (The Plexxie Global Company Pte Ltd) and its email domain (plexxie.com).

- search_for_email(company_name, domain)



If the job description page does not contain any email address in the Application Instructions section, this function will be used. The code will type some keywords, including the company name and its domain extracted beforehand, to search for the potential emails on Google. For example, “IGG Singapore Pte Ltd igg.com contact email”.

After entering the keywords, there are two ways to fetch the potential email addresses:

1. In the search results, the code will directly extract every email shown, such as [“cooperation@igg.com”](mailto:cooperation@igg.com).
2. The code will select links that contain words such as "Contact", "About", or "Privacy" in the title. For example, the link containing the title “Contact US - GAMERS AT

HEART” will be saved. If no relevant links are found, the code will automatically select and store the first three result links.

After storing the links, each of them will be visited by the code to search for any email addresses on the page. All found emails will be checked to ensure they belong to the correct domain, such as “@igg.com”. Only the emails that fulfill this requirement will be stored.

- choose_best_email(email_addresses_list)

Now that we have a list of found emails, this function helps select the most appropriate one.

There are three situations to determine the best email:

1. If the list contains only one found email, this email will be selected as the best email.
2. A list of around 15 job-related keywords is defined within this function, ranging from most relevant to least relevant, such as “careers”, “jobs”, and “hr”. If the list has multiple found emails, the code will choose the most relevant one based on these predefined keywords. For example, “jobs.sg@igg.com” would be chosen due to the presence of the keyword “jobs”.
3. If no email matches any of the keywords, you may need to manually select the most appropriate email from the list of the found emails.

- save_to_excel(data, filename='Joblist updated to 2025.1.XX.xlsx')

	A	B	C	D	E	F	G
1	Company	Position	E-mail	E-mail List	Domain	Date	Link

This function saves every extracted record into an Excel file. The file includes the above seven columns, which are populated with the extracted data.

If the Excel file “Joblist updated to 2025.1.XX.xlsx” does not exist on your computer, the code will create a new one. Otherwise, it will append new records to the last row of the existing file.

- process_batch(start_page, end_page)

This is the most important function, utilizing the six previously introduced functions to carry out the entire fetching process. The following outlines the two main scrapping steps, which will loop until all posts on a specific page are handled.

«	1	...	64	65	66	67	68	...	120	»
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Company	Title	Location	Period	Post
I Was Here Pte Ltd iwasherewhere.com	Project Manager (Art & Design)	SG	Immediate Start, For At Least 3 Months Full/Perm Free/Proj Intern/IS	18 Nov
Nanyang Beverage Pte Ltd nanyangbev.com	Creative Marketing Intern	SG	From Mar 2025, For At Least 6 Months Intern/IS	18 Nov

Step 1: Visit a specific page (e.g., page 66).

- For each post, check if it is still open for applications.
- If it is open, check if the position title contains the word “intern” (e.g., “Creative Marketing Intern”).
- If the title contains “intern”, obtain the post date (for instance, 18 Nov).

Step 2: If the post fulfils all the requirements mentioned in Step 1, visit the job description page. (For this example, use the link to “Creative Marketing Intern” with the function `create_chrome_driver(headless=True)`.)

On the job description page:

- Obtain the company name and its domain using the function `extract_company_name(detail_soup)`.
- Check if there is any email provided by the company under the Application Instructions section using the function `extract_contact_info(detail_soup)`.
 - If an email is found, it will be stored.
 - If no email is available, search for email addresses on Google using the function `search_for_email(company_name, domain)`.
- If emails are found on the Internet, select the most relevant email using the function `choose_best_email(email_addresses_list)`.
- Finally, append the extracted data to the list and save it into the Excel file using the function `save_to_excel(data, filename='Joblist updated to 2025.1.XX.xlsx')`.

3. Run the code

To execute the code, click on “Execute Cell”. By default, the code is configured to fetch data from pages 66 and 67: `batch_data = process_batch(66, 67)`.

The entire scraping process takes approximately 6 minutes to complete.

If your network is fast and your computer has sufficient RAM and CPU power, you may adjust the page load time. The line `time.sleep(4)` indicates a 4-second wait for the page to load.

- If you reduce this to `time.sleep(1)`, the fetching process may complete in roughly 4 minutes.

If you want to rerun this code cell, it is advisable to **restart the kernel** to avoid any interruptions during the next fetching process.

4. Understand limitations of the code and possible solutions

There are three notable limitations in the code:

1. It is possible that sometimes, even if the code can search for email addresses on the Internet, it fails to choose the most relevant one as the best email (the yellow cell) because no email matches the predefined keywords.

117	INITIA Group SG	Art Intern	['business@initia.sg', 'jsmith@initia.sg']	initia.sg	26 Dec	https
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2. There is a chance that no relevant email (the green cell) is found on Google. This can be because the corresponding company does not disclose its emails on the Internet or does not have an official website or social media accounts.

268	Molor Pro Pte. Ltd.	HR Intern	[]	molorsg.com	13 Dec	https
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3. In some rare cases, the domain provided on the job description page does not belong to the company (the blue cells), which can lead to incorrect email scraping.

299	Jest Media	Events Coordinator (Intern)	careers@eventions.com.sg	['info@eventions.com.sg', 'careers@eventions.com.sg']	eventions.com.sg	13 Dec	https
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If we encounter these situations, it may be necessary to choose, search for, or double-check the emails ourselves. If you do not want to work manually on this, you may choose to simply delete all the records that contain nothing in the E-mail column by running the **second code cell**. While this method can save you time, it may result in the loss of part of the records in the job list.